```
import matplotlib.pyplot as plt

from tensorflow.keras.datasets import cifar10
import cv2
```

```
image = "/content/image.jpg"
image = cv2.imread(image)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

Task 2

```
def show_image(image):
  plt.figure(figsize=(5, 5))
  plt.imshow(image)
  plt.axis('off')
  plt.show()
```

```
show_image(image)
print(f"No of channels : {image.shape[2]}")
```

```
No of channels : 3
```

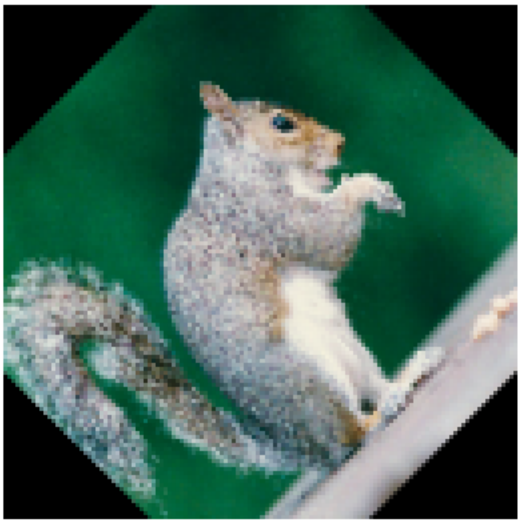
task 3

```
grey_image = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)
fig, ax = plt.subplots(1, 2, figsize=(10, 6))
ax[0].imshow(image)
ax[0].axis('off')

ax[1].imshow(grey_image, cmap='gray')
ax[1].axis('off')

plt.show()
```

Task 4

```
resized = cv2.resize(image, (128, 128))
show_image(resized)

(h, w) = resized.shape[:2]

center = (w // 2, h // 2)
```

```
# Rotation matrix (45 degrees)
M = cv2.getRotationMatrix2D(center, 45, 1.0)

# Rotate image
rotated = cv2.warpAffine(resized, M, (w, h))

show_image(rotated)
```



```
horizontal_flip = cv2.flip(resized, 1)
show_image(horizontal_flip)
```

```
gaussian = cv2.GaussianBlur(resized, (5, 5), 0)

median = cv2.medianBlur(resized, 5)
```

```
fig, ax = plt.subplots(1, 2, figsize=(10, 6))

ax[0].imshow(gaussian)
ax[0].set_title("Gaussian Blur")
ax[0].axis("off")

ax[1].imshow(median)
ax[1].set_title("Median Blur")
ax[1].axis("off")

plt.tight_layout()
plt.show()
```